

SIMATS Online Programming Lab

JAVA PROGRAMMING

AL NAJEEB A

192424325RB

CO2-Assignment

QUESTIONS

Q1

Bank Token Queue Management System

10 marks

Q2

ATM PIN Security Validator

10 marks

Q3

Warehouse Damaged Item Removal

10 marks

3/3 answered

Q1 Bank Token Queue Management System

10 marks

A bank serves customers based on token order. Using Queue:

Add customer tokens using offer()

Serve the next customer using poll()

Display the upcoming customer using peek()

Your Code

```
1 import java.util.LinkedList;
2 import java.util.Queue;
3 public class Main {
4     public static void main(String[] args) {
5         Queue<Integer> q = new LinkedList<>();
6         q.offer(1);
7         q.offer(2);
8         q.offer(3);
9         System.out.println("Queue: " + q);
10        System.out.println("Next customer: " + q.peek());
11
12        System.out.println("Served: " + q.poll());
13        System.out.println("Queue now: " + q);
14        System.out.println("Next customer: " + q.peek());
15    }
16 }
```

Input

stdin input

Output

Queue: [1, 2, 3]
Next customer: 1
Served: 1
Queue now: [2, 3]
Next customer: 2

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3/3 answered

Q2 ATM PIN Security Validator

10 marks

A nationalized bank receives ATM PINs from customers through a kiosk. Develop a Java program that accepts the PIN as a string and performs the following operations:

Remove unwanted spaces using trim()

Validate that the PIN contains exactly 4 digits using matches()

Display the PIN length using length()

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String pin = sc.nextLine();
7         pin = pin.trim();
8         boolean ok = pin.matches("\\d{4}");
9         System.out.println("PIN Length: " + pin.length());
10        if (ok) System.out.println("Valid");
11        else System.out.println("Invalid");
12    }
13 }
14 }
```

Input

1212
1221

Output

PIN Length: 4
Valid

Run

Save

Not secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=739



SIMATS

ENGINEERING

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Q1

Bank Token Queue Management System

10 marks

✓

Q2

ATM PIN Security Validator

10 marks

✓

Q3

Warehouse Damaged Item Removal

10 marks

✓

3/3 answered

A warehouse stores item names in an inventory list. Damaged items must be removed safely while traversing the collection.

Requirements:

Store inventory items in an ArrayList

Traverse the list using an Iterator

Remove items marked as Damaged and display the remaining inventory

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         ArrayList<String> inventory = new ArrayList<String>();
5         inventory.add("Soap - Good");
6         inventory.add("Rice - Damaged");
7         inventory.add("Milk - Good");
8         inventory.add("Wheat - Damaged");
9         inventory.add("Sugar - Good");
10
11         Iterator<String> it = inventory.iterator();
12         while (it.hasNext()) {
13             String item = it.next();
14             if (item.contains("Damaged")) {
15                 it.remove();
16             }
17         }
18         System.out.println("Remaining Inventory:");
19         for (int i = 0; i < inventory.size(); i++) {
20             System.out.println(inventory.get(i));
21         }
22     }
23 }
```

Input

stdin input

Output

Remaining Inventory:
Soap - Good
Rice - Damaged
Milk - Good
Wheat - Damaged
Sugar - Good